

Amir TRABELSI, PhD

Robotics R&D Engineer

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Robotics R&D Engineer with over 5 years of experience in the development of multi-robot systems, specialized in minimally invasive surgical applications. PhD in Mechanical Engineering with expertise in optimal design and kinematic redundancy management. Proven track record in mathematical modeling, hardware-software integration, and high-complexity robotic platform development.

PROFESSIONAL EXPERIENCE

Medical Robotics R&D Engineer | LS2N Laboratory - RoMaS Team, Nantes, France Nov. 2024 – Present

- Design and development of a robotic assistant for percutaneous interventions.
- Implementation of real-time control algorithms for macro-mini robotic architectures.
- Development of EtherCAT communication interfaces and APIs for multi-system synchronization.
- **Tech Stack:** ROS 2, Real-time control, EtherCAT, UDP, Robotic Simulation, Mechatronic Design, Git.
- Medical Robotics Researcher** | Institut Pprime - CoBRA Team, Poitiers, France July 2021 – Oct. 2024
- Developed a multi-robot system for minimally invasive surgery.
- Optimized robot placement to enhance kinematic and dynamic performance.
- Designed control laws for multi-robot coordination within shared workspaces.
- Implemented kinematic redundancy management strategies based on real-time optimization.
- Conducted experimental validation and simulation via digital twins.
- **Tech Stack:** ROS, Matlab, SolidWorks, Numerical Optimization, KUKA iiwa, Franka Emika.

EDUCATION

PhD in Robotics | Institut Pprime - CoBRA Team, University of Poitiers, France 2021 – 2025

- Recipient of the Eiffel Excellence Scholarship.
- Thesis: Optimal design and redundancy management of a multi-robot platform for minimally invasive surgery.

M.Sc. in Mechanical and Systems Engineering | ENISO, Tunisia 2018 – 2021

- Graduated Top of Class.
- Thesis: Development of a high-performance simulator for Franka Emika cobots in Matlab/Simulink for medical applications.

Engineering Degree in Mechatronics - Robotics Specialty | ENISO, Tunisia 2017 – 2020

- Thesis: Inertial parameter identification of a medical tool for a cobot equipped with a 6-axis force sensor.

TECHNICAL SKILLS

- **Robotics & Control:** Kinematic/Dynamic modeling, Mechatronic design, Real-time control, Multi-robot coordination, Redundancy management, Numerical optimization, Digital Twins.
- **Software & Simulation:** SolidWorks (CSWP/CSWA certified), CATIA V5, Blender, Gazebo, MuJoCo, Qualisys, MATLAB/Simulink (Advanced), LabVIEW.
- **Programming:** ROS/ROS 2, C/C++, Python.
- **Languages:** English (Fluent), French (Bilingual), Arabic (Native).

SCIENTIFIC PUBLICATIONS (SELECTED)

- **A. Trabelsi**, et al. "Robot Base Placement and Tool Mounting Optimization based on Capability Map." *Robotica*, 2024.
- **A. Trabelsi**, et al. "Design and Implementation of a Digital Twin for a Multi-robot Surgical Platform." *ISRM*, 2024.
- **A. Trabelsi**, et al. "Optimal Multi-robot Placement Based on Capability Map for Medical Applications." *RAAD2022*, 2022.
- **A. Trabelsi**, et al. "Development of a Franka Emika cobot simulator platform (CSP) dedicated to medical applications." *RAAD2021*, 2021.

ACTIVITIES & LEADERSHIP

- **Active Member** | ARBL Nantes Aeromodelling Club (2024–Present).
- **President** | ENISO-TEAM Robotics Club (2018–2019).
- **Event Organizer** | ENISO Smart Challenge V5 (2018–2019)